STAPLES STEM JOURNAL



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staples high school september 2023

The Staples STEM Journal provides an outlet for individuals to share their STEM interests with the Staples community and aims to broaden public interest and knowledge in these fields.

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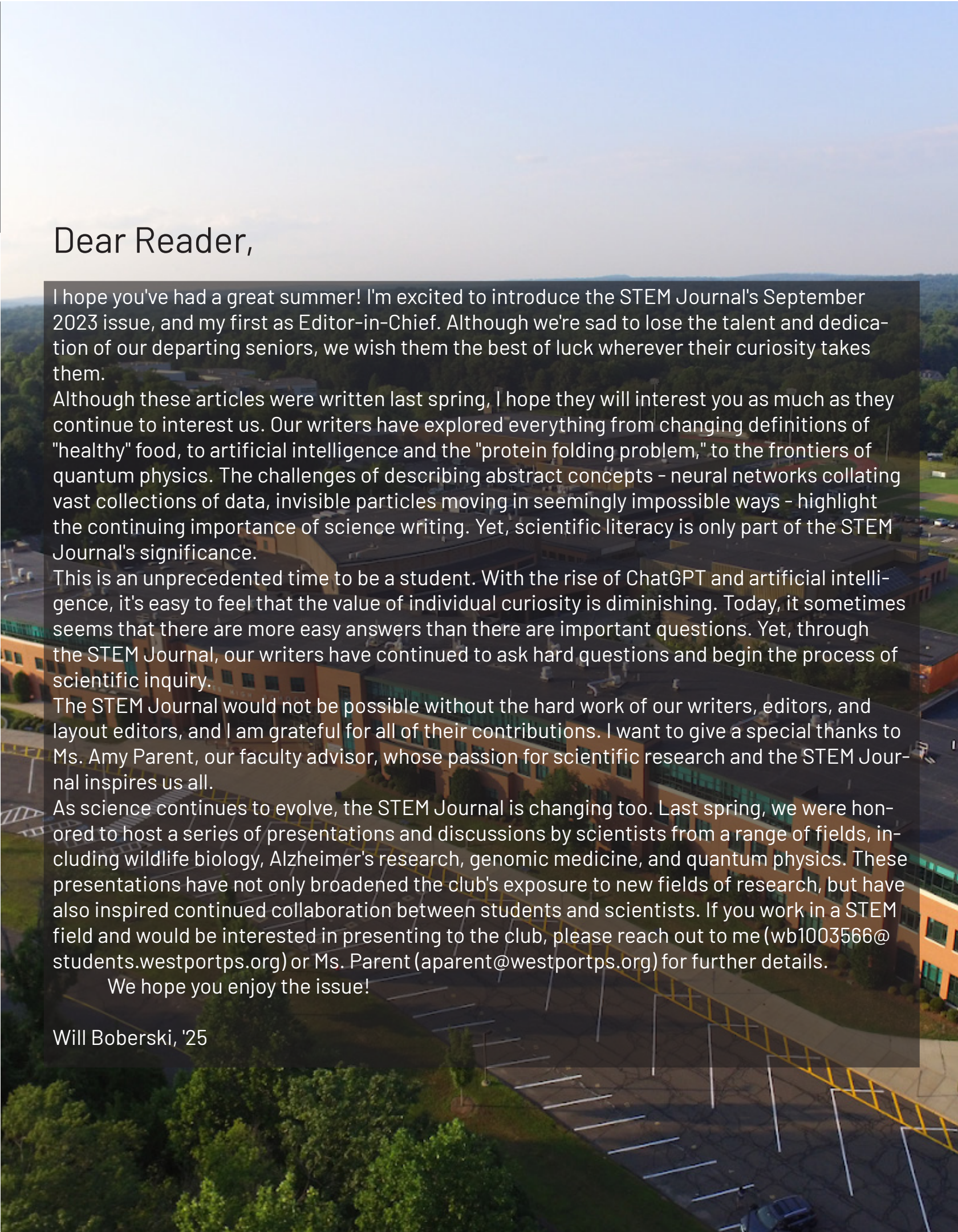
@Staples STEM Journal

We are thankful for the support of our excellent advisor, Ms. Amy Parent.

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Dear Reader,

I hope you've had a great summer! I'm excited to introduce the STEM Journal's September 2023 issue, and my first as Editor-in-Chief. Although we're sad to lose the talent and dedication of our departing seniors, we wish them the best of luck wherever their curiosity takes them.

Although these articles were written last spring, I hope they will interest you as much as they continue to interest us. Our writers have explored everything from changing definitions of "healthy" food, to artificial intelligence and the "protein folding problem," to the frontiers of quantum physics. The challenges of describing abstract concepts - neural networks collating vast collections of data, invisible particles moving in seemingly impossible ways - highlight the continuing importance of science writing. Yet, scientific literacy is only part of the STEM Journal's significance.

This is an unprecedented time to be a student. With the rise of ChatGPT and artificial intelligence, it's easy to feel that the value of individual curiosity is diminishing. Today, it sometimes seems that there are more easy answers than there are important questions. Yet, through the STEM Journal, our writers have continued to ask hard questions and begin the process of scientific inquiry.

The STEM Journal would not be possible without the hard work of our writers, editors, and layout editors, and I am grateful for all of their contributions. I want to give a special thanks to Ms. Amy Parent, our faculty advisor, whose passion for scientific research and the STEM Journal inspires us all.

As science continues to evolve, the STEM Journal is changing too. Last spring, we were honored to host a series of presentations and discussions by scientists from a range of fields, including wildlife biology, Alzheimer's research, genomic medicine, and quantum physics. These presentations have not only broadened the club's exposure to new fields of research, but have also inspired continued collaboration between students and scientists. If you work in a STEM field and would be interested in presenting to the club, please reach out to me (wb1003566@students.westportps.org) or Ms. Parent (aparent@westportps.org) for further details.

We hope you enjoy the issue!

Will Boberski, '25



1. Sugary Yogurt is Out, Salmon is In. Redefining Healthy Food; The Impacts of the FDA's Updated Definition.

Nolan Francis, '26

As of 2022, 39% of American adults have a gym membership; similarly, over the past 10 years, the fitness industry has grown from $50.2 million to $65.1 million (Kolmar 2022). Simply put, American society has become more health conscious and, consequently, more focused on what we put in our bodies. But, what defines healthy food?

Background

The Food and Drug Administration (FDA) is the government organization responsible for protecting public health in the U.S. by regulating human drugs, biological products, and food (FDA, 2022). The FDA approves foods for human consumption and may officially label foods as healthy if they meet certain requirements. However, for the past few years, these requirements have been criticized for their inaccuracy and arbitrary specificity.

As of 2022, 5% of all packaged foods are labeled as "healthy" by the FDA. The definition of healthy foods set in 1994 enabled manufacturers to label products as "healthy" if they contained limited amounts of total fat, saturated fat, cholesterol, and sodium. In addition, healthy foods were required to provide 10% of the daily recommended amount of vitamins A, and C, calcium, iron, protein, or dietary fibers. Fresh fruits, vegetables, game meats, and fish had slightly different requirements (FDA, 2022).

The definition of healthy was renewed in 2016 to reflect a modern understanding of nutrition. The FDA allowed some foods to contain higher amounts of total fats and others to only meet the requirement of containing at least 10% of one's daily total recommended amount of vitamin D or potassium. However, the FDA had no criteria for added sugars, an omission they recognize as inconsistent with today's science (FDA, 2022).

"The old rule was really outdated—you could create any kind of Frankenstein food that met the nutrient criteria and label it as healthy," said Dr. Dariush Mozffarian, a cardiologist and professor of nutrition at the Tufts Friedman School of Nutrition Science and Policy in Boston, "[updating the definition] is a major advance" (Blum 2022).

On September 28, 2022, the FDA unveiled its new proposal, modernizing the definition of healthy foods so that it adheres to contemporary health science.

What does FDA's updated proposal mean?

Proposed at the White House Conference on National Hunger, Nutrition, and Health, the updated definition limits added sugars to 2.5 grams per serving (varies based on food, ex. fruits), caps sodium at 230 grams per serving, and limits saturated fat based-on the type of food.

Under the old definition, a six-ounce serving of yogurt containing ten grams or more of sugar would be labeled as healthy, but a frozen dinner containing salmon, green beans, and brown rice would not be labeled as healthy because it contained over four ounces of total fats. The updated definition fixes this, encouraging healthy eating by prioritizing major food groups such as vegetables, fruits, grains, dairy and proteins, and certain oils. Healthy food needs to contain at least one of the major food groups and abide by the proposed limits for saturated fats, sodium, and added sugars. Raw fruits and whole vegetables qualify automatically (NYT 2022); this definition clearly makes more sense.

"Those criteria will eliminate vast swaths of the supermarket from being eligible for the healthy logo," as stated by Marion Nestle, a professor of nutrition, food studies, and public health at New York University. Most sugary cereals, granola bars, highly sweetened yogurts, and white loaves of bread, which might currently qualify as healthy would be eliminated under the new regulations. Nuts, avocados, seeds, fatty fish (like salmon), and certain oils, which do not currently qualify, could earn the distinction under the new guidelines. The new definition focuses on how a food fits into a healthy dietary pattern overall as opposed to individual nutrients. For instance, salmon, which is currently not labeled as healthy because it is high in fats, would earn the healthy distinction because it is rich in beneficial omega-3 fatty acids and protein and low in saturated fats and cholesterol.

Next Steps

"Typically, after the FDA proposes a rule, the agency seeks commentary from outside health experts and the general public before it goes into effect," said Dr. Peter Lurie, executive director and president of the Center for Science in the Public Interest, "the process can take one year or more." While he applauded some aspects of the proposed update, especially the limit on added sugars, Dr. Lurie stressed that there is a fundamental problem with labels; added sugar amount would remain voluntary as with the current label. Consumers might erroneously think that any foods without labels are automatically unhealthy. "It's not really helpful in that respect," he said, "it allows the industry to decide what to convey to the consumer, as opposed to providing the consumer with what they would clearly want" (Blum 2022).

In response, Dr. Lurie and others in the nutrition field are pushing for standardized mandatory nutrition labels placed on the front of food packages, which the FDA is currently looking into. The FDA hopes the updated definition will help consumers make better dietary decisions to help lower the incidence of "diet-related chronic diseases" like cardiovascular diseases and Type 2 diabetes. According to the FDA, over 80% of the people living in the U.S. aren't getting enough vegetables, fruit, and dairy in their diets. "There's been so much mixed messaging on what's healthy and what's not," Dr. Rajagopal said, "the average consumer just doesn't have a baseline" (Blum 2022).

Ultimately, the FDA's redefinition of the term "healthy" is long overdue. The modernized definition can potentially benefit consumers because it provides a clear baseline as to what nutritional foods should be part of a balanced diet. While this is not the only solution to improving national nutrition, it does have the potential to benefit American public health greatly.

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2. The Future Role of *Hericium erinaceus* as a Therapeutic Agent of Alzheimer's Disease:

Abraham Lobsenz '25

Introduction

Pharmacognosy is in fashion: millions of natural products citing miraculous medicinal qualities have taken the supermarket and restaurant industries by storm. While some stress the potency of these products, their bioactivity is often more limited than marketed, and subjective to one's holistic health practices. That said, there remains a basis for exploring the medicinal properties of natural substances, some of which outperform even synthetic pharmaceutical drugs. Since diet "is an important modifiable risk factor for Alzheimer's Disease (AD) as it is able to modulate structural brain connectivity, cause positive changes in brain function and behavior, and help regulate cognition and emotion," there has been an increasing focus surrounding neuroprotective natural products in particular (Li, 2018). *Hericium erinaceus*, also known as Lion's mane mushroom (LMM), is one such example that has proven to exhibit therapeutic activities related to nerve and brain health, offering an alternative to synthetic AD drugs with an admittedly unreliable track record. For instance, N-methyl-D-aspartate receptor antagonists, drugs that may mitigate stroke and traumatic brain injuries, have proven efficient in AD rodent models yet superfluous in human trials. Levodopa, the primary treatment for Parkinson's, effectively crosses the Blood Brain Barrier (BBB) to reach impaired neurons, but long-term use can elicit negative consequences. Indeed, only 1 drug out of 244 tested in 413 clinical trials between 2002 and 2012 was approved for AD treatment, and it didn't even work (Li, 2018). So, besides its natural origin, how exactly is LMM any different?

Classification of *Hericium erinaceus* history and cultivation:

LMM is "an ideal culinary mushroom", or rather, fungi, widely available in most health-oriented supermarkets and major online retailers (Li, 2018)(Cascadia, 2021). The taste has been described by most as reminiscent of seafoods, sweet, rich, and somewhat savory. While its popularity has soared to unprecedented levels today, its roots can be traced back to ancient civilizations as a "medicinal ally" (Cascadia, 2021). LMM appears as a single clump of pure white dangling spines (See Figure 1), which browns progressively as it ages. These 'spines' appear as hairs that develop a fibrous and stringy texture when ingested, but this can be avoided through cooking, which also helps to break down chitin cellular walls to promote optimal nutrient absorption. Many names have referred to this curiously shaped mushroom, including Monkey's Head, Bearded Tooth, Satyr's Beard, Bearded Hedgehog, and Pom-Pom Mushroom (Cascadia, 2021). Initially, LMM was cultivated in Asia serving as a traditional Chinese medicine, where it was prepared as a tonic with the capacity to improve all

of the known internal organs at the time. Religiously, it was thought to prevent degradation of "Qi", or life force. Primarily conserved for royalty, it was valued among the monks for its supposed cognitive benefits. LMM's wild cultivation was first reported in 1988, its health benefits alluring enough to promote large-scale industrial crops on indoor substrates, the process that is employed today (Cascadia, 2021). Throughout North America and Europe, LMM fruits in the late summer and early fall. In addition to a host of nutritional advantages, such as Vitamin D and antioxidant contribution, LMM also displays immune support, anti-tumor, and of course, cognitive activity (Cascadia, 2021).

Exploring the Biochemical Composition of *Hericium erinaceus*



Figure 1`:Lions Mane Mushrooms (Kirkpatrick, 2021)

The primary bioactive compounds within LMM are Erinacines and Hericenones, the first present largely in the mycelium of LMM and the latter occupying the fruiting body. Erinacines are part of a structurally diverse class of natural compounds only found in mushrooms, cyathin diterpenoids. All show some level of biological activity mostly as stimulators of Nerve Growth Factor (NGF) synthesis, a protein that could treat the mild stages of neurodegeneration in AD (Li, 2018). Of the 15 erinacines (named alphabetically - See Figure 2) discovered to date, ones A-I demonstrate NGF secretion and reduction of beta-amyloid plaques (a prominent protein in interneuron pathway degradation), while E also demonstrates mitigation of neuropathic pain. While these erinacines all offer possible AD pharmaceutical applications, there remains no direct evidence that they can effectively cross the Blood Brain Barrier (BBB), an urgent basis for future studies (Li, 2018). In spite of this, compounds from natural substances typically offer unique BBB solubility as well as limited side effects. Hericenones have failed to demonstrate promotion of NGF in 1321N1 human astrocytoma cells, ruling them out almost completely for any meaningful treatment of AD (Li, 2018).

The Efficacy of Erinacines for AD treatment as Contributed By in Vitro/Vivo Studies

Theoretically wise at least, LMM's consultants, Erinacines and Hericenones, are both seemingly chemically unviable or controversial. However, the key appeal of these compounds are the few but incredibly promising in vitro and in vivo studies that establish their neuroprotective properties. Indeed, despite failing to produce measurable changes in NGF synthesis, Hericenones demonstrated improvement of cognitive decline among 50-80 year Japanese patients, though this is mostly the extent of their neuropathic benefits (Li, 2018). Conversely, Erinacines, Erinacine A in particular, have proven to exhibit much broader implications on measurable factors of AD.

Thus far, Erinacine A is the only one that has the capacity to match in vitro experimentation to in vivo results (See Figure 3)(Li, 2021). One study at erinacine A dosage of 8 mg/kg body weight enhanced the amount of NGF and demonstrated increased neuronal survival and behavioral maintenance. Another study showed that erinacine A could reduce beta-amyloid plaques by increasing their degradation. After a certain point in AD progression, the evidence shows that accumulation of beta-amyloid is a runaway chain (Rodriguez, 2022). A heightened presence of these plaques also extends hyperphosphorylated tau protein seeding. Thus, it follows that beta-amyloid mitigation (perhaps via LMM supplementation) should take research precedence in treatment options. Yet, another

study of Erinacine A in transgenic mice (19 mg/g administered) reduced the activation of plaque-associated microglia and astrocytes and the expression of insulin-degrading enzymes, alternative agents in AD origination and progression (Li, 2018). As if this single compound didn't excel enough already, the efficacy of erinacine A in downregulating certain neuro degrading genes also upregulated genes involved in the prevention of malignant tumors! It must be noted that the intricacy of AD mechanisms has presented extensive issues in extrapolating these studies to clinical situations, but preclinical evaluations have seen improvements in Parkinson's, Alzheimer's, and even depression (Li, 2018). In the very few human studies thus far, results have been likewise promising. After 49 weeks of Erinacine A supplementation, a significant decrease in Cognitive Abilities Screening Instrument score was noted in the placebo group in a study of propably-AD-impaired people, while a significant improvement in Mini-Mental State Examination score was observed in the supplemented group. Moreover, "only the placebo group observed significantly lowered biomarkers such as calcium, albumin, apolipoprotein E4, hemoglobin, and brain-derived neurotrophic factor and significantly elevated alpha1-antichymotrypsin and amyloid-beta peptide 1–40 over the study period" (Li, 2021). However, preliminary research on the direct taupathic effects of H. erinaceus bore less fruit, predicting "no improvements in spatial memory nor activities of daily living" of taupathic mice. The study did, however, corroborate others in regards to beta-amyloid degradation, and illustrated anxiolytic effects in the mice (Rodriguez, 2022). According to researchers, future studies could focus on "analyzing…information about the mushroom's effects on tau levels, oxidation, astrocyte activity and inflammation, fiber growth and survival, and maintenance of nerve cells," as well as determining

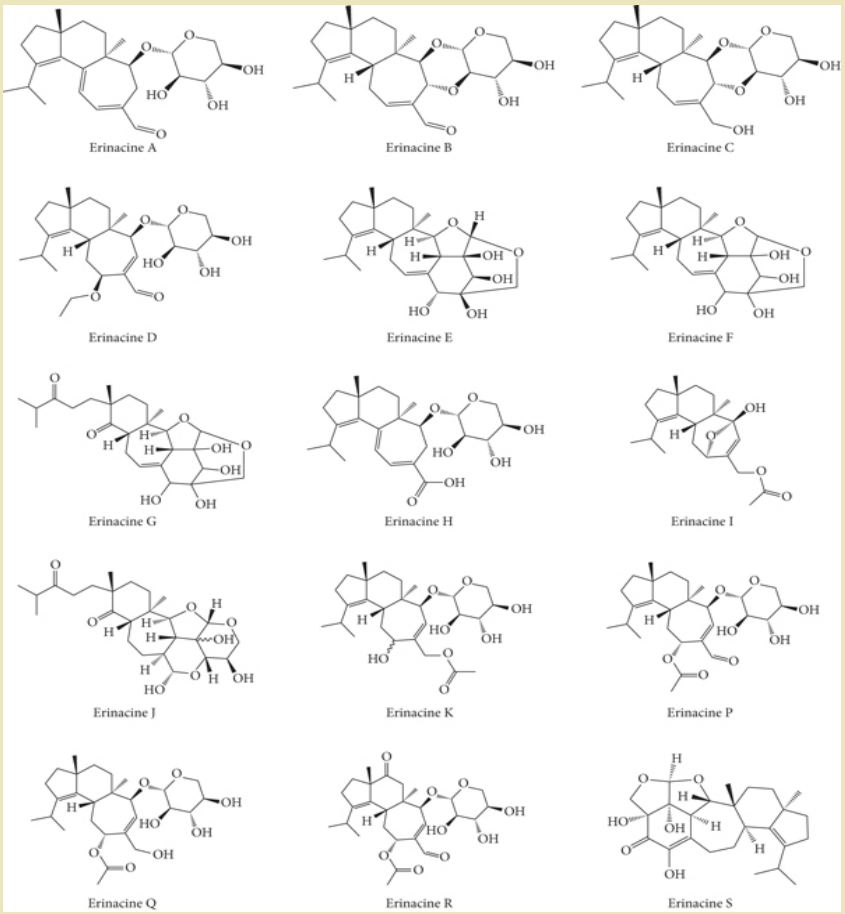


Figure 2:Neurohealth Properties of Hericium erinaceus *Mycelia Enriched with Erinacines (Li et al., 2018.)*

erinacine solubility in the BBB (Rodriguez, 2022). Overall, current findings label Erinacine A as a promising therapeutic option in delaying and alleviating AD progression.

Evaluation of Erinacine A Extraction and Consequential Market Viability

So, Erinacine A is a probable agent in AD pharmacognosy, potentially destined to replace synthetic drugs as a therapy. There is one complication with this dynamic: due to the micro quantities of this compound in LMMs mycelium, an economically viable extraction process must be developed. The most common approaches are enhancing erinacine production in *Hericium erinaceus* via a fermentation process, employing bioreactors to increase mycelia yield (biosynthesis as opposed to chemical) and increasing concentrations of bioactive metabolites, thus expanding the mushroom's medicinal functionality and potential for drug extraction (Li, 2018). This strategy has been adopted by some bioreactors, producing erinacine A and C in a reactor consisting of glucose, casein peptone, sodium chloride, zinc sulfate, and Potassium dihydrate phosphate at a pH of 4.5. This process produced 192 mg erincine A after 8 days of cultivation. Certain optimization of these parameters could help to maximize the biosynthesis process of most-promising erinacine A (Li, 2018). Scaling up this biosynthesis is also crucial to future extraction. There has been one successful case of commercial production already, its highest accumulation being 5mg/g erinacine A with 20-ton fermentors after 12 days (Li, 2018). With this commercialization in mind, the first erinacine A-enriched *H. erinaceus* mycelium product was introduced to the market in 2015 in Taiwan (Li, 2021). Considering the natural identity of this substance, isolation from *Hericium erinaceus* serves as an ideal quality control method.

Erinacine A	*In vitro*	1 mM	Induced 250.1 ± 36.2 pg/ml NGF synthesis
	In vivo	30 mg/kg body weight/day	(1) Reduced amyloid burden by 38.1 ± 19.7% (2) Increased IDE levels by 141.1 ± 63.7%
	In vivo	1 mg/kg body weight/day	Inhibited DLD-1 tumor growth by 66%
	In vivo	30 mg/kg body weight/day	(1) Reduced both the size and number of amyloid plaques (2) Increased IDE levels by 303.5% (3) Recovered from impairments in burrowing, nesting, and Morris water maze tasks

Figure 3: Neurohealth Properties of Hericium erinaceus Mycelia Enriched with Erinacines (Li et al., 2018.)

Conclusion

LMM has upheld medicinal applicability through the centuries, and it seems as though it will once again prosper, this time as a future platform from which the effects of more extensively studied erinacine A can be augmented. In the meantime, LMM supplements, including a minute dosage of erinacines, can be found on amazon for the comparatively low price of $14.39/ounce (Amazon, 2023). This is not to totally discount the other erinacines, and even hericenones, as more studies are desperately needed to establish the true mechanisms behind their neuroprotective and bioactive properties, or lack thereof. While the supplementation/health industries have exaggerated the capabilities of pharmacognosy before, and will continue to mislead their consumers to turn a profit, they may have pegged LMM, or erinacine A-enhanced LMM at least, just right.

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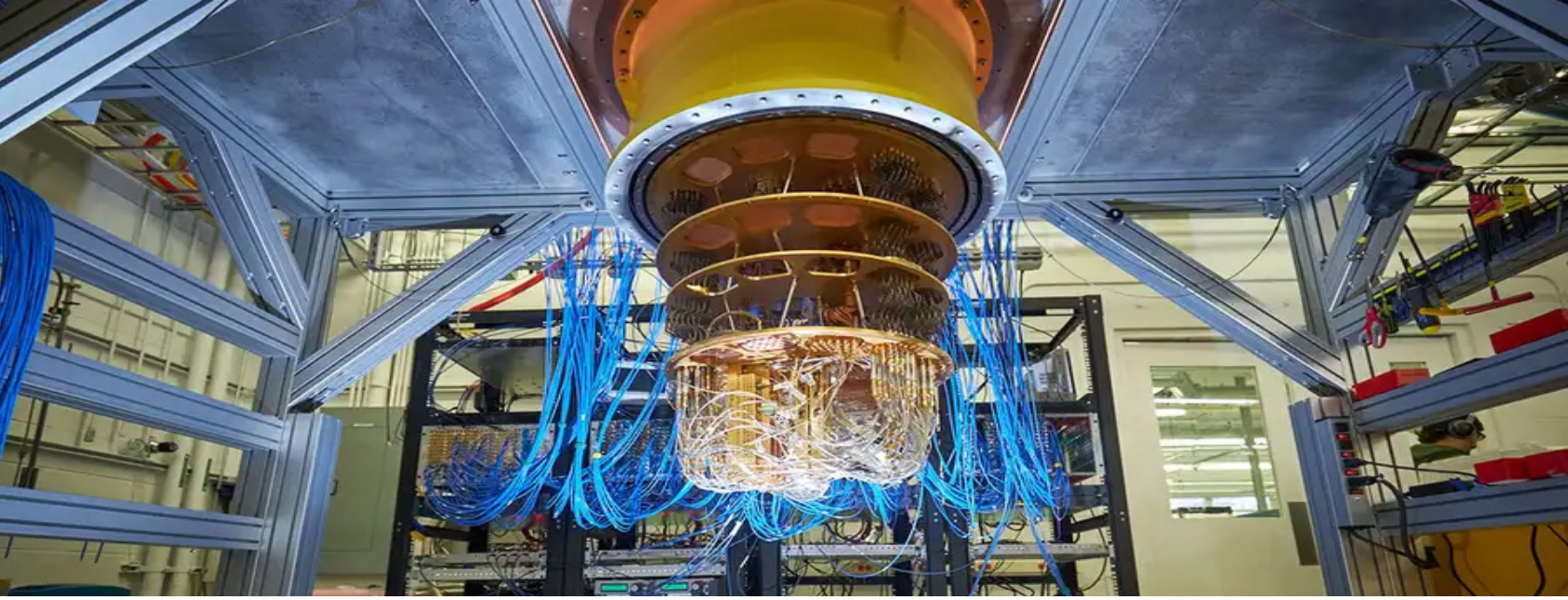
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3. A Brief History of Quantum Computing and Predictions for the Future.

Andrew Rebello '25

Introduction

Quantum computers. Although they may sound complex, these special computers are, at their core, quite easy to understand. They are based on simple quantum physics principles, making them quite different from classical computers, or the computers used in everyday life.

History of Quantum Computers

First brainstormed in the 1980s by American physicist Richard Feynmann and Russian mathematician Yuri Manin, quantum computers were designed to address a major problem in physics (Hudek et al., 2022).

The computers used today are too inefficient to simulate nature, namely atoms and molecules. Simulating systems of even a few dozen interacting particles requires more computing power than any conventional computer can provide over thousands of years!

To illustrate this with a very simple example, let's consider two particles interacting with each other. The particles have one property that matters to us: charge. At the subatomic level, calculations must be made for the electrostatic force, collisions, etc., but we will ignore that for now. Let us make the further assumption that charge has 2 possible values. Currently, there are only 4 possible configurations for the system of charged particles. If we add a third particle, there are 2^3=8 possible combinations of charges. Add until you get 40 particles and you have 2^{40}(over a trillion) possible combinations. Just to store the information about charge alone would take about 130 GB, according to Microsoft Azure experts. To add another particle, an extra 130 GB is needed (Hudek et al., 2022). We aren't even using memory to make calculations, so evidently simulating quantum configurations on a classical computer would take a very long time and is simply infeasible. And that's the concern: scientists aren't interested in 2 particles, but rather systems of many, many particles. Therefore, the idea behind quantum computers was: if we cannot apply computers to nature, we should apply nature to computers. In other words, instead of using conventional computers to understand quantum physics, what if we applied the principles of quantum physics to computers and used it to our advantage?

Understanding the Basics

There are two principles of quantum mechanics that lend themselves to the power of quantum computing: superposition and entanglement.
To illustrate superposition, there is a famous thought experiment: Schrödinger's Cat. Schrödinger imagined placing a cat in a sealed box with a device for one hour. However, the device would have a 50% chance to kill the cat and a 50% chance to leave the cat unharmed. Before we open the box, guess whether the cat is dead or alive. Although intuition might prompt us to answer either 'dead' or 'alive', Schrödinger said that the cat was actually dead and alive at the same time. He claimed that the cat was in a state of superposition between the two states. Particles at the quantum level are similar: until we observe them for say, spin, they are in a state of superposition between up spin and down spin. But—analogous to opening the box—the second the particle is measured, it collapses into one of the two possible states (TED-Ed, "*Schrödinger's cat*").

Schrödinger's Cat also serves as a good analogy for Quantum Entanglement. Imagine two of those boxes with identical cats and the same 50/50 odds. When we close both boxes for another hour, there are four possible combinations: both cats are alive, both cats are dead, the first cat is alive and the other is dead, and vice versa. Physicists in the 20th century ruled out the possibility of the cats both being alive or both cats being dead. This means that by knowing whether one cat is alive or dead, you know whether the other is alive or dead. This is quantum entanglement: the properties of a particle can be figured out just by knowing the properties of its entangled particle (TED-Ed, "*What*").

		Cat 1	
		Alive	Dead
Cat 2	Alive	25%	25%
	Dead	25%	25%

Table based on intuition:

		Cat 1	
		Alive	Dead
Cat 2	Alive	x	50%
	Dead	50%	x

Table based on the reality of quantum mechanics:

In computer science, the way to determine how an algorithm scales is referred to as Big-O Notation. An algorithm with Big-O, O(n) scales linearly, so doubling the number of elements would double the time taken to run the program. An $O(n^2)$ algorithm scales quadratically, so doubling the elements would quadruple the run time. The example we used earlier with the participles would be Big-O $O(2^n)$, exponential time. But a Big-O in exponential time scales terribly, so much so that computer scientists classify problems with such algorithms as unreasonable. Although the computer could hypothetically solve it using such algorithms, the time it takes to run becomes more unreasonable as inputs scale.

Qubit vs. Bit

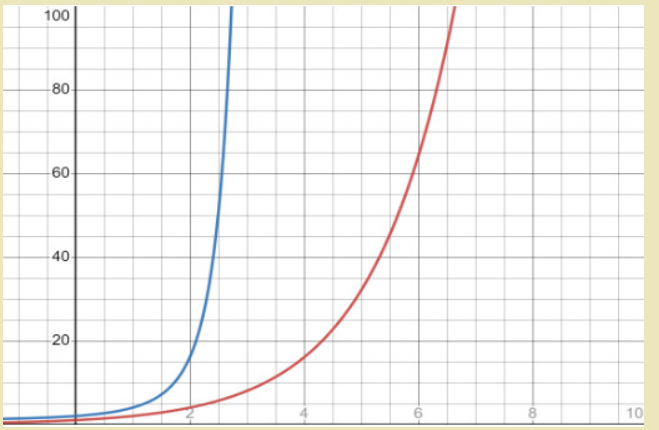
The keyword in computers is bit: bits store information in 0s and 1s. With 2 bits, you can represent 4 numbers in binary: 00(0), 01(1),10(2), and 11(3). So if you wanted to perform an operation on all 4 numbers, you would have to perform the operation on every possible combination via a brute force approach.

Quantum computers have quantum bits, qubits for short. Superposition differentiates these qubits from their classical counterparts: the qubit doesn't necessarily have to be 0 or 1, but can be in both states at once. The instant the qubit is measured, however, it collapses into a definite state: 0 or 1. This simple principle is a game changer: if you have four bits, there are 2^4=16 possible combinations, and you can only use one. But if you had four qubits in superposition, they could stimulate all 16 combinations at the same time. 20 qubits can already run a million values in parallel (Kurzgesagt – In a Nutshell, 2015). Scientists are hoping that quantum computers with such qubits would hope to solve many problems originally deemed unreasonable for classical computers.

Entanglement is another game changer: it can make computations faster. Financial Review explains that, "In a quantum computer, entanglement is used as a sort of computational multiplier for qubits. As you entangle more and more qubits together, the ability of the system to make calculations grows not in a linear fashion, but exponentially" (Davidson, 2019).

Classical vs. Quantum: Using the right tool for the job

So will quantum computers replace classical computers? Although the answer is unknown, quantum computers won't be on commercial markets for a long time. The first reason is simple: quantum computers need to be kept really cold. The more energy the qubit has, the more unstable it is. Therefore, quantum computer refrigerators keep them a fraction of a Kelvin above absolute zero to minimize the energy in the system — therefore minimizing the probability of a qubit unintentionally changing its state. This cooling power is unfeasible for the average consumer, hence why they may not replace classical computers in the foreseeable future. The second reason is that quantum computers are probabilistic, meaning they may output different results every time the code is run based on probability. This is another difference between quantum computers and normal deterministic computers, which provide the same result no matter how many times the program is run. Although this may not initially seem helpful, it is a fair tradeoff to access quantum computers' powerful ability to do unreasonable problems in reasonable time. Principles like superposition and entanglement don't lend themselves well to the firm rules that are required to run software like file management, which must store information consistently. According to BuiltIn, quantum computing is so "fundamentally different from classical computing that it will take time to develop, deploy and reap the benefits of the technology" (Joury, 2021). In other words, quantum computers aren't X times more powerful, but a completely different game that we are still learning how to play.



A graph of $2^{2^{\wedge}x}$(blue) compared to the growth of 2^x(red) to reference how fast Neven's "doubly exponential" growth is compared to standard exponential growth.

Humans prefer straightforward answers that can be obtained by classical computers, but we also need answers to complex problems. Those could be solved with quantum computers. So, both classical and quantum computers will likely stick around for decades to come.

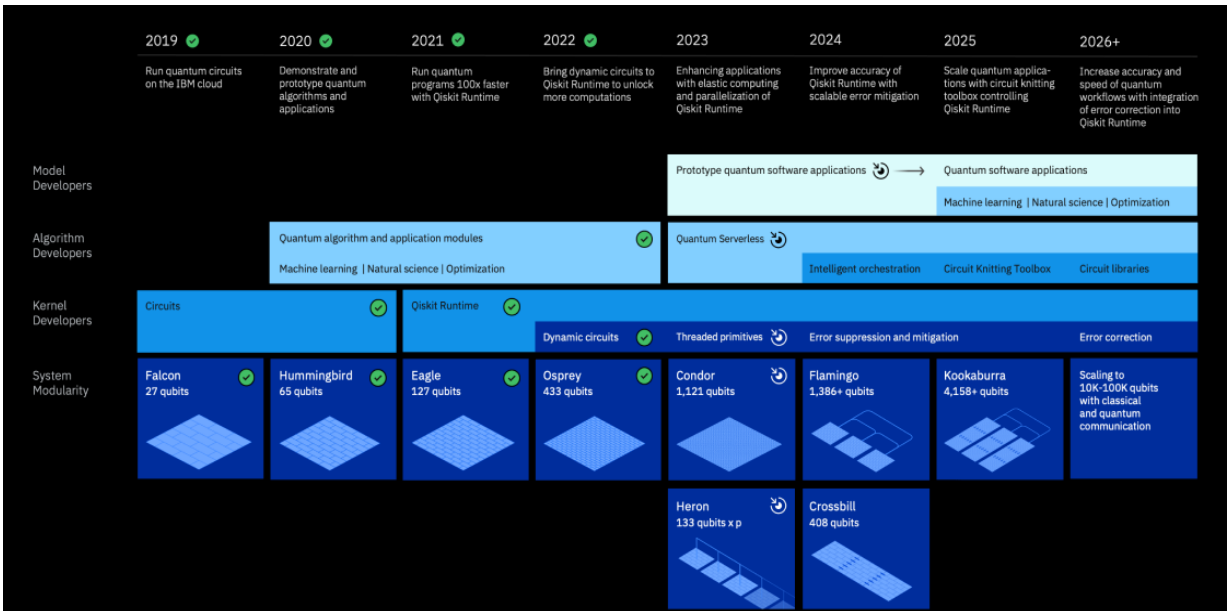
Conclusion

In conclusion, quantum computers will definitely be a rising force to solve complex problems, but will not take over classical supremacy. Nonetheless, quantum growth is incredible.

Moore's Law is the famous observation that the number of transistors in a classical computer chip doubles about every two years, an exponential rate (like 2^x). But Google's Quantum AI lab director Hartmut Neven characterizes quantum computers as gaining computational power relative to classical ones at a "doubly exponential" rate (like 2^{2^x}). (Hartnett) Whether Neven's Law will hold against the test of time remains to be seen: Andrew Childs, the co-director of the Joint Center for Quantum Information and Computer Science at the University of Maryland, asserts that, "When looking at all the moving parts, including improvements on the classical and quantum sides, it's hard for me to say it's doubly exponential." But regardless of characterizing the exact rate, quantum technology is significantly improving nonetheless.

Finally, a main goal scientists are trying to solve right now to improve the science is minimizing external interference on qubit. Qubit states are easily influenced by interference from heat, magnetic fields, noise, etc. That said, all of this contributes to increasing error probability. Minimizing this would minimize error and lead to more accurate results a greater portion of the time.

Additionally, scientists work to decrease decoherence time. For reference, decoherence is the process by which a quantum particle, in this case a qubit, interacts with its environment and collapses into a set state — this would inference with quantum algorithms. For algorithms that utilize quantum



Above: IBM's "ambitious" timeline for progressing quantum computing in the coming years. They have met these annual goals so far, and are on track to continue that pattern. They have already built Osprey, the largest quantum computer to date, and are leading the field. This year, they are breaking a 1,000 qubit milestone with Condor (IBM, 2019).

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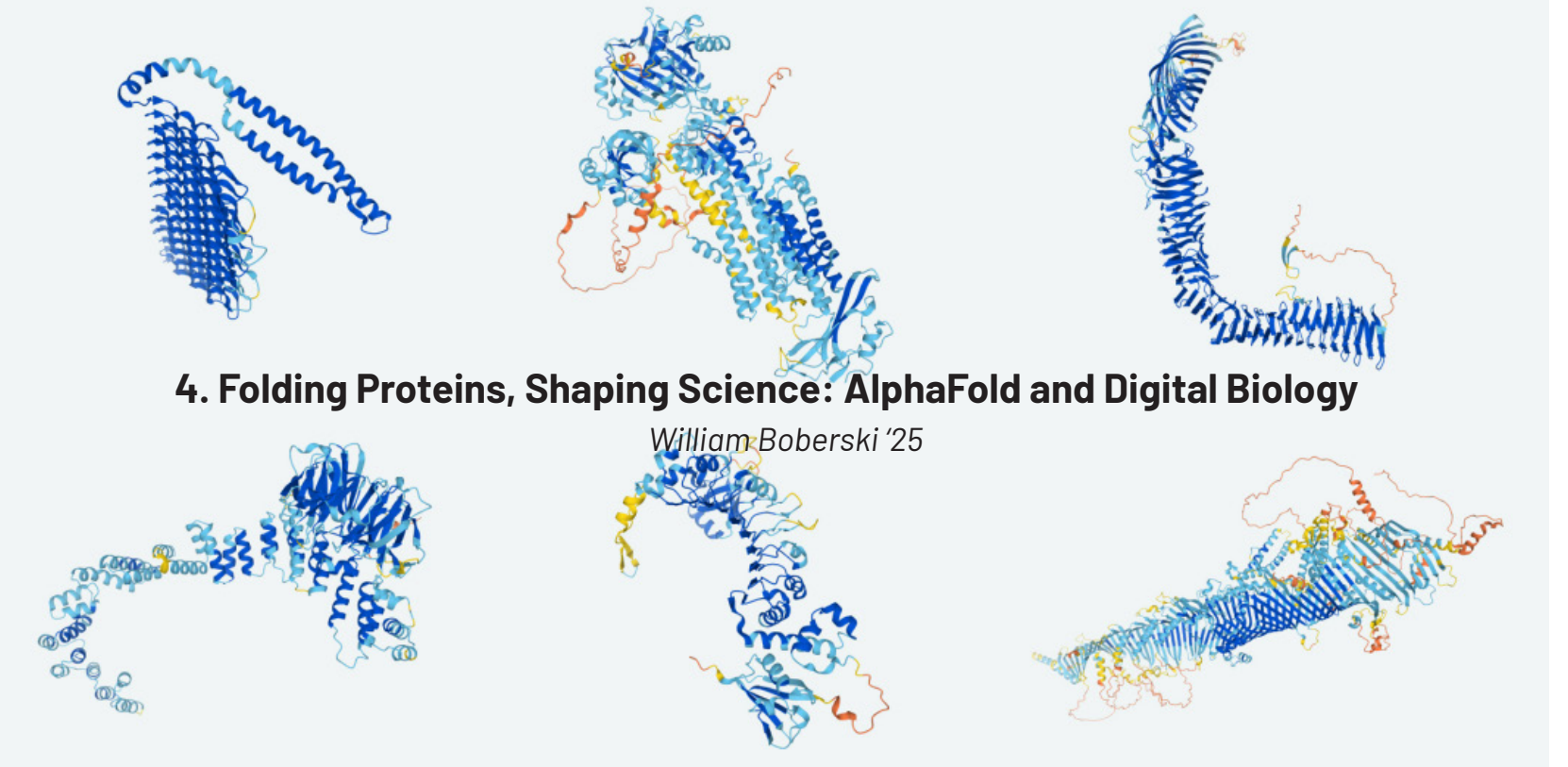
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4. Folding Proteins, Shaping Science: AlphaFold and Digital Biology

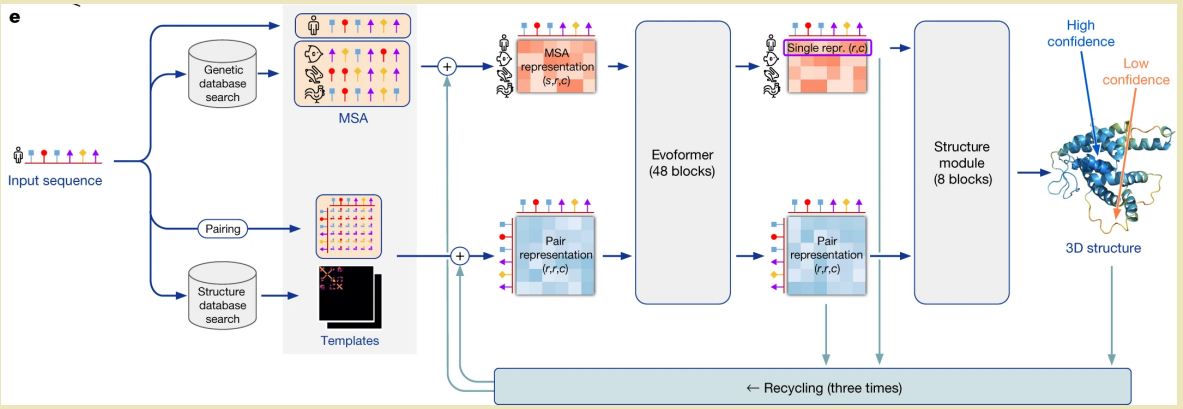
William Boberski '25

At this moment, a new researcher is examining the shapes of countless proteins. Proteins are made up of one or more long chains of amino acids, which take complex shapes once they are complete. The question of how proteins get their shape is known as the protein-folding problem, and usually involves painstaking, time-consuming laboratory imaging. Yet, this researcher has never set foot in a lab, nor have they ever worked with an actual protein. The researcher is AlphaFold, an artificial intelligence (AI) tool developed by the AI research laboratory DeepMind (DeepMind).

Neural Networks and Protein Structure Prediction

AlphaFold is an open source, neural network-based model that uses deep learning to predict rotein structures (Jumper, 2021). Neural networks, whose architecture is based on the neural structures of the brain, contain multiple algorithms to find patterns in large data sets (Chen, 2022). More specifically, deep learning networks contain as many as 150 "hidden layers" of algorithms, working together to recognize patterns directly from data. Traditional machine learning systems, by contrast, sort data based on programmed features and classifiers (MathWorks). AlphaFold was "trained" on the sequences and structures of around 100,000 known proteins (DeepMind) from the Protein Data Bank (PDB). A network was also used to predict the structures of around 350,000 additional proteins; the high-confidence predictions helped supplement the PDB data to improve AlphaFold's accuracy (Jumper, 2021).

The precise shape of a protein is determined based on the properties of its components. For example, amino acids each have different physical properties, such as being hydrophobic or hydrophilic, while the atoms that make up these monomers also have different chemical properties. Covalent and hydrogen bonds between amino acids contort the chain into a specific shape. In the human body, polypeptide chains fold into proteins when mediated by chaperone molecules and enzymes (Cooper, 2000). AlphaFold, however, can predict protein shape based on only the properties of the constituent amino acids and their atoms. AlphaFold accurately predicts protein structure down to the atomic level; the model also estimates its own reliability, color coding areas of high confidence in blue and areas of low confidence in orange (Jumper, 2021). By applying biochemical principles, the AlphaFold neural network has "learned" to predict the shape of nearly every protein known to science.



Above: Figure 1 illustrates the architecture of the AlphaFold neural network. The blocks labeled "Evoformer" and "structure module" each contain different operations to analyze and interpret data based on different features; the sequence is "recycled" through these operations three times to improve the accuracy of the predictions (Jumper, 2021).

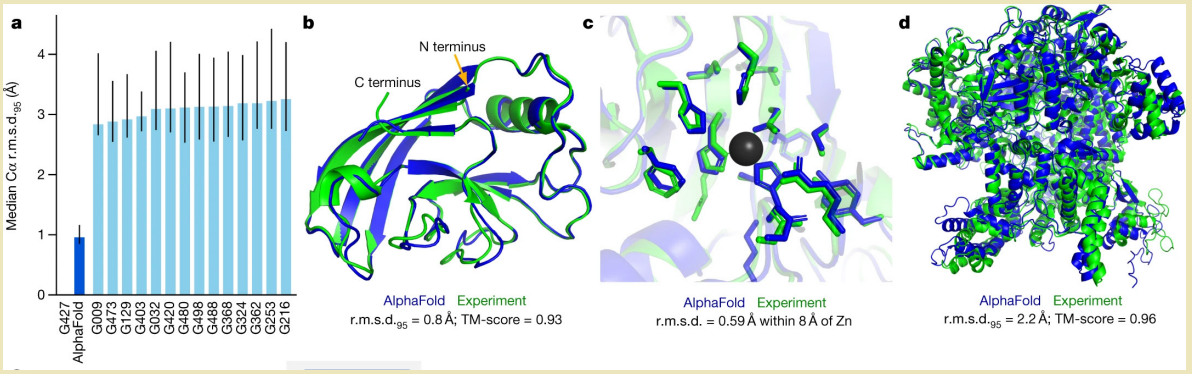


Figure 2 Above: AlphaFold's structure predictions (in blue) are highly accurate to the structures obtained through experimental methods (in green), even on the atomic level (see Figure C) (Jumper, 2021).

Solving the Protein Folding Problem

DeepMind's development team knew that AlphaFold could predict protein structure. However, could it solve the protein folding problem? Since 1994, the international Critical Assessment of Protein Structure Prediction (CASP) competition has challenged 100 groups of researchers to accurately predict the shapes of 100 unknown proteins, in the hope of finding a solution to the protein folding problem. Competitors were assessed based on the Global Distance Test (GDT), which compares the prediction to a "ground truth," or known shape. During CASP 13 in 2018, AlphaFold allowed the DeepMind team to make dramatic progress towards a solution for the protein folding prediction (DeepMind, 2020). The results from CASP 14 in 2020 were even more decisive. Using a new version of AlphaFold, the DeepMind team predicted the shapes of 2/3 of the target proteins with a GDT of over 90, thus achieving a solution to the protein

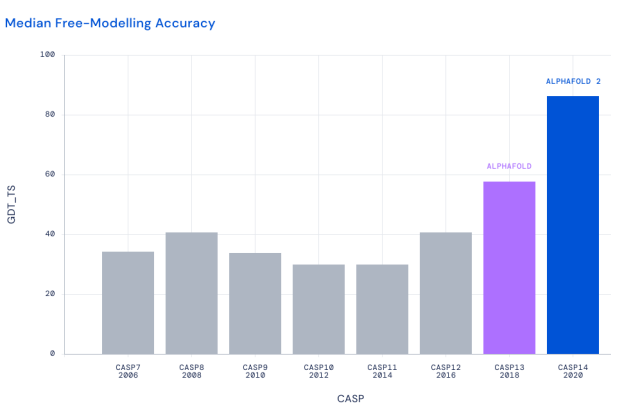


Figure 4: AlphaFold has significantly improved the accuracy of protein folding predictions, measured here based on the GDT scores of the top 5 teams participating in CASP competitions (Hassabis, 2021).

folding problem (Noble, 2020).

In partnership with EMBL's European Bioinformatics Institute (EMBL-EBI), AlphaFold has created the open source AlphaFold Protein Structure Database. The database contains nearly every protein known to science - over 200 million - as well as the predicted structures of the entire human proteome (Hassabis, 2022). After decades of experimental work, structures had been determined for only 17% of the total human proteins; the PDB contained entries for 35% of human proteins, but most were fragmentary. AlphaFold, however, has predicted the structures of 98.5% of human proteins; 58% of the predictions are considered to be confident, while 36% of all predictions are considered very confident (Tunyasuvunakool, 2021).

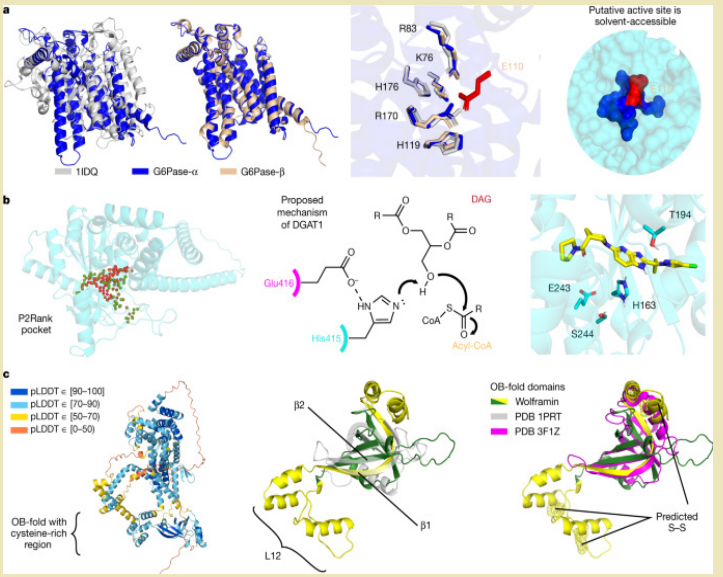


Figure 5: AlphaFold has predicted the active sites of two glucose-forming enzymes (see Figure A), a binding pocket for a molecule used in body-fat synthesis (see Figure B), and a protein, wolframin, whose mutations could cause a rare disorder (see Figure C) (Tunyasuvunakool, 2021).

Diseases, Enzymes and Molecular Machinery

AlphaFold in Practice

By helping scientists understand protein folding, AlphaFold provides insight on how protein structure determines biological function. Structural data can help determine "the druggability of a given protein target, and [...] the design of small-molecule drugs that will bind to it." Furthermore, drugs developed for specific protein structures may lead to "fewer toxic side effects." Protein structure data may also help determine which mutation variants are responsible for disease, identify drug targets in infectious agents, and improve vaccine and antibody design (Thornton, 2021).

Scientists around the world are already using AlphaFold to accelerate their research, as illustrated by the following examples. Nuclear pore complexes (NPCs) are some of the most complex molecular machines in human cells, comprising more than 1,000 protein subunits of 30 different types. Building on existing experimental data, researchers used AlphaFold to predict the shapes of the remaining protein subunits. The AlphaFold results corresponded with scientific expectations for protein shapes, and have helped to generate a model of 2/3 of the NPC (Unfolded, "AlphaFold"). Structural biology researchers have even used AlphaFold in conjunction with cryo-electron tomography in order to better understand the architecture of human NPCs (Mosalaganti, 2021).

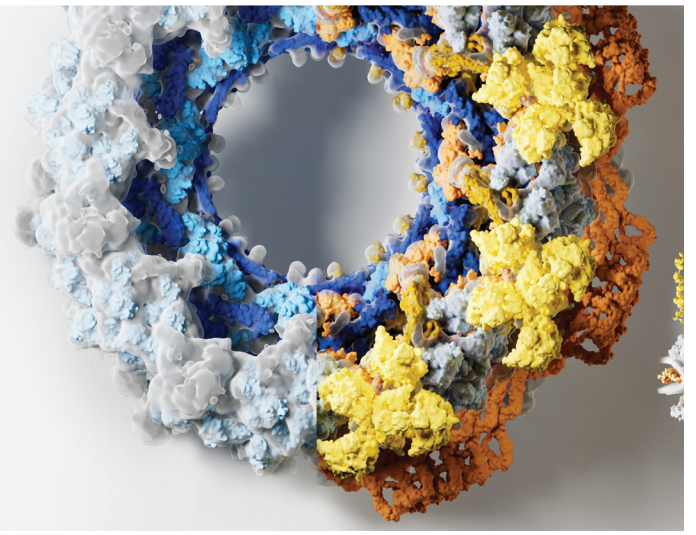
AlphaFold has helped the Drugs for Neglected Diseases Initiative (DNDi) improve a chemical that could be used to treat leishmaniasis. Carried by parasites and spread by sand fly bites, cutaneous leishmaniasis causes skin lesions, while visceral leishmaniasis can be fatal. Although DNDi had developed a potential drug for leishmaniasis, researchers struggled to improve it using traditional methods of random chemical changes. The drug affected an enzyme in the parasite used to make phospholipids. The enzyme, however, was embedded in the cell membrane, making it difficult to experimentally determine its structure. Using AlphaFold, DNDi analyzed the drug-enzyme interaction in order to develop a potential treatment for leishmaniasis (Unfolded, "The race").

AlphaFold may provide new solutions to address increasing antibiotic resistance. For example, the Sousa Lab at the University of Colorado, Boulder aims to develop drugs that target bacteria's antibiotic resistance mechanisms. For 10 years, these scientists have worked to image antibiotic resistant enzymes in bacterial cell membranes. Membrane proteins are difficult to study using X-ray crystallography, and this method doesn't definitively determine the protein's shape. By combining AlphaFold's structure predictions with X-ray crystallography data, these scientists hope to accelerate research into antibiotic resistance (DeepMind, "Meet Marcelo"). AlphaFold may also help researchers better understand the structures of viruses and other pathogens, thus speeding up drug discovery and vaccine development.

AlphaFold may also help address the ecological impacts of materials like plastic. As plastic pollution increases, enzymes to break down plastic may help mitigate environmental degradation. Understanding the function of an enzyme, however, depends on the ability to visualize its structure. By quickly and accurately determining protein shapes, AlphaFold enables researchers to explore new plastic recycling solutions (DeepMind, "Meet John").

Digital Biology: AlphaFold and the Future of Science

AlphaFold is part of a revolution in structural biology, heralding the beginning of what some have termed "digital biology." AI can now be used to predict structural models that once took months or years of research, enabling the creation of large datasets to compare protein shapes. Biology has entered "an era of structural abundance," and accelerated exploration is sure to follow (Hassabis, 2022). Demis Hassabis, the director of DeepMind, has founded Isomorphic Labs, which aims to



"redefine drug discovery with the power of artificial intelligence." Isomorphic Labs approaches biology as the study of "information processing system[s]," which shares the "basic underlying structure" of "computer science" (Isomorphic Labs). Although the goals and projects of Isomorphic Labs remains abstract, the future of structural biology and drug discovery are promising, and have already started to become a reality. AlphaFold isn't perfect; it struggles to predict multi-domain and flexible proteins, and doesn't model ligands, which help to mediate protein interaction (Thornton, 2021). After all, AI hasn't replaced biologists, and it never will. Instead, the possibilities for future exploration in structural biology have never been broader or more exciting.

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5. The Avian Flu Epizootic: Human Impacts

Paige Foran '26

Avian influenza, also called the avian or bird flu, has appeared in its worst outbreak to date that began in 2021. Predominantly seen in poultry, domestic, and wild birds, this infectious disease is believed to have originated in Europe and Asia between ducks. As the virus has spread and variants of it have developed, such as the H5N1 variant that concerns the current epizootic, it has reached infection in other animals such as seals, wild dogs, otters, foxes, and more. Concerns have been raised amongst scientists as they consider the alarming potential of the virus to mutate and spread more easily to humans, giving rise to the next epidemic.

Since the beginning of the outbreak in October of 2021, there have been almost 42 million cases of the disease in domestic and wild birds. Roughly 15 million domestic birds (including poultry) have died from the disease, and over 193 million have been slaughtered. What's particularly unique about this epizootic is that the most wild birds have also been killed by the virus than ever before, for it has impacted over 80 different species. In even more devastating news, it is nearly impossible to control and contain the spread of H5N1 between wild birds; it has now reached the point at which individuals must take measures to prevent the spread of the disease to humans. It has been advised to disinfect or eliminate bird feeders and bird baths and to wear heavier-duty masks when handling animals, yet there is only so much that these strategies can achieve.

Scientists are still stumped as to why this outbreak is so severe: there has been nothing like it up to date. The leading hypothesis at the moment is that the virus has undergone an incredibly effective mutation that is able to more easily jump from organism to organism and/or last in a given environment for longer. Furthermore, it is doubted whether or not there will be complete resolution to the outbreak. "As the virus now has infected many wild bird species, it becomes unlikely that it will disappear again from the bird population," says Nancy Beerens, an avian flu expert at Wageningen Bioveterinary Research in the Netherlands.

Though it has not yet reached the point of human vaccination, professionals have considered options for animal vaccinations: they are wondering how effective a potential vaccine could be and if it is even possible financially, and they had generally come to the conclusion that, from past experi-

ence with COVID-19, the cost to success ration would not be worth the effort. Thus, to public knowledge, there is no current, global work on a major poultry vaccine. China has decided to begin vaccinating their poultry, but the meat and eggs from these vaccinated flocks are not sold internationally. In regards to human infections, there have been very few rare cases of the avian flu. Only 10 cases have been documented, and just one of those cases was said to be found in the United States. While this may seem rather encouraging, influenzas have the remarkable ability to mutate and change rapidly, so medical professionals must be wary of the fugacious situation at hand. Also, the disease is still severe, for a young girl in Cambodia died of this flu in February of 2023. Such a tragedy reminds society that even a predominantly zoonotic virus should be taken seriously with infection rates at this level; spillovers from animals to humans are almost always possible.

Overall, the seemingly drastic numbers of the avian influenza infections have a certain reassurance in them, for humans do not necessarily have to be concerned about contracting this disease easily. However, reports of cases are certainly not unheard of, and scientists must be on the lookout for variants of the virus, like the H5N1 one now, that are more efficient and devastating than those recorded. But, it is also important to take away the idea that millions of birds are severely impacted by this; prices of eggs have been driven up, and the future likely holds other uncertain economic impacts. So, being aware of this situation with the potential to escalate to drastic levels can play a major role in ending this epizootic before it's too late,



Avian flu is especially volatile for birds used for domestic poultry, threatening human health as well as the poultry industry (Pearce, 2023).

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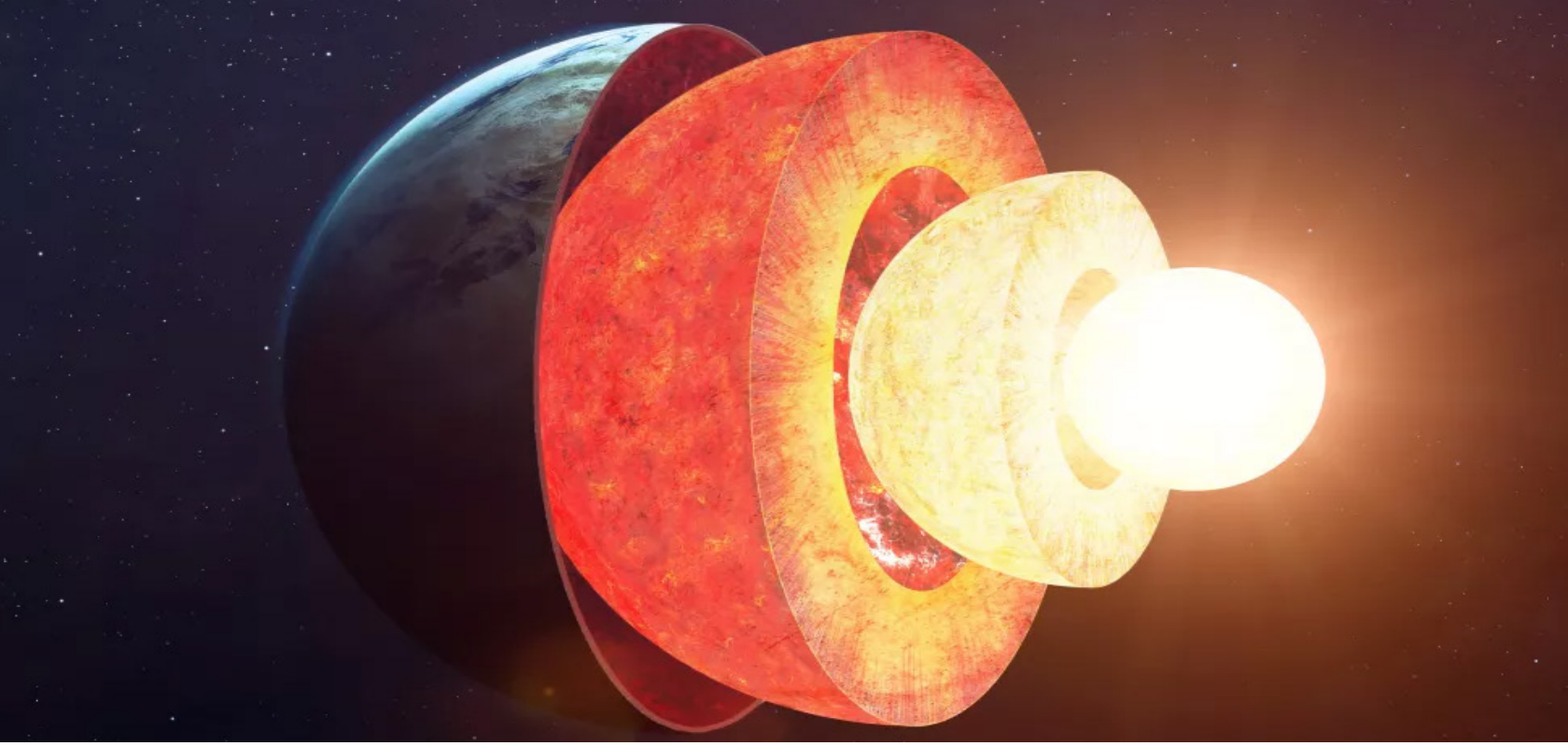
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6. Recent Discoveries Concerning Earth's Core

Zachary Gottlieb '25

From the ocean tides to raising mountains, the Earth performs giant geological processes every day. These phenomena are all intrinsically connected to one another, exerting their influences in a plethora of ways, and working together to sustain life in diverse conditions.

The Earth's core plays a vital role in tectonic plate movement and generating the magnetic field. This article will give a background on the layers of the Earth, and explore two recent papers concerning the structure and rotation of the inner core at the center.

Core Concepts

Before going over the discoveries, it first seems reasonable to give a "brief" introduction to the structure of the Earth. The physical structure of the Earth is comparable to an onion with four (maybe five, this will be discussed later) distinct layers: The crust, mantle, outer core, and inner core. Unfortunately for scientists, this "onion" is exceptionally difficult to conceptualize, for "except for the thin crust we live on, Earth's structure is intangible deep beneath our feet" (Kuthunur, 2023).

On the surface of the planet, the lithosphere contains the continental and oceanic crusts, along with the upper part of the mantle. Beneath this - but still in the mantle - is the asthenosphere, where convection currents push at the tectonic plates. Further down, the transition zone gate separates the upper and lower mantle, with the lower mantle being much hotter than the former (Buis, 2021).

Beneath the indomitable depths of the mantle, with a chemical composition of mostly iron and nickel, the Earth's core is roughly the size of Mars (Kolirin, 2023). Like the mantle, the core is also divided, with two main domains that are starkly different. Separated from the planet's surface by 3,200 miles of miles of rock, the solid inner core is encased in what *Scientific American*'s Stephanie Pappas calls a "liquid cocoon". This "cocoon" is a molten-metal outer core in which "the inner core sits suspended like a ball bearing" (Pappas, 2023). Scientists speculate the inner core is 1,520 miles wide (in diameter), and might be about as hot as the sun's surface (Andrews, 2023).

Seismic Probing and the Innermost Inner Core

But how do scientists know these measurements about the core? Hrvoje Tkalčić, a geophysicist and seismologist at the Australian National University, answers this question in an email to Space.com. The "inner core is notoriously difficult to probe by seismic waves" he says, but by studying how seismic waves caused by large earthquakes get distorted as they go through the core can help to *unearth* the mysteries of the Earth's depths (Kuthunur, 2023). Tkalčić was a co-author of a recently released study in March that found evidence of an "innermost inner core". This isn't a new discovery, as the existence of another layer was first theorized in 2002 (Sullivan, 2023).

First authors of this study, Thanh-Son Pham and Tkalčić looked at data from large earthquakes of the past. As the waves reverberate, they lose energy with each pass through the planet. The echoing waves had faint signals which the researchers combined to detect the rebounding waves (Pappas, 2023). Pham said that due to the recent installments of new seismic sensors around the globe, it's increasingly possible to detect these weak seismic signals (Kuthunur, 2023). These waves rippled across the Earth's diameter five times, the highest reflection rate ever recorded. As the seismic waves pass through this region, the speed is slowed depending on the angle the wave hits the core. They concluded that the iron crystals are organized differently between the two layers of the core. The innermost inner core exhibits an "anisotropic" phenomenon, which allows a material to possess different properties in different directions. They estimate that this layer of the core is 800 miles in diameter, and while the two layers are of similar composition, they have different crystal structures (Patel, 2023). This is just the most recent in a string of observations, most of the Earth's core still remains a mystery. Pham says, "we may know more about the surface of other, distant celestial bodies than the deep interior of our planet" (Sullivan, 2023).

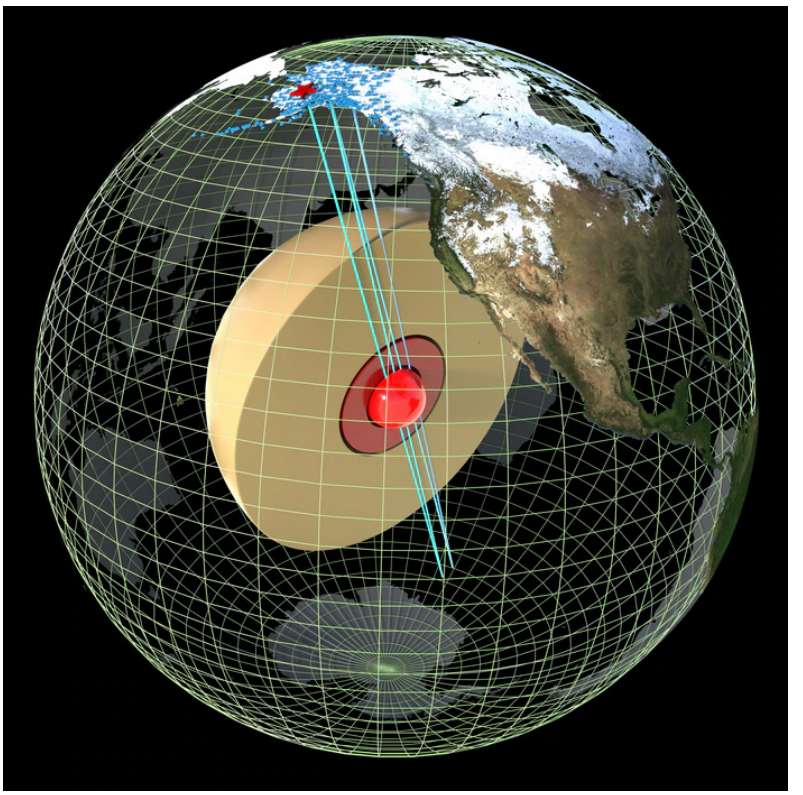


Diagram depicts an earthquake in Alaska causing seismic waves to penetrate the Earth's innermost inner core (Patel, 2023).

A Slowing Core and the Importance of a Magnetic Field

In late January, a study analyzing seismic wave data was published by researchers at Peking University in China. It concluded that the spin of the inner core's rotation had stopped in 2009, and then proceeded to reverse direction. Authors of this paper, Yi Yang and Xiaodong Song, believe that this is a part of a 7-decade cycle, with the last rotation change occurring in the early 1970s (Hurst, 2023). The researchers compared two nearly identical earthquakes - in the same location with the same magnitude - only separated by time, and noticed subtle differences in the rebounded waves (Pappas, 2023).

Based on their calculations, they conclude the reason for this switch ultimately comes down to two reasons: the Earth's magnetic field and gravitational forces acting on the core (Kolirin, 2023). The Earth's magnetic field - called the magnetosphere - is generated in the outer core, in a mechanism called "geodynamo", where the heat convection in the fluid outer core causes the iron to make an electrical current. The magnetosphere is essential to protecting life from cosmic radiation and solar winds (Buis, 2021). As for the gravity, the composition of the mantle and inner core are thought to be heterogeneous (mixed material). Consequently, in a process called "gravitational coupling", the gravity between these domains forces the inner core into a position of a gravitational equilibrium, further inducing spin (Kolirin, 2023). Each of these processes contribute to the rotation of the inner core.

Seismic studies have shown that "each year the core expands by about a millimeter, as some of the molten iron in the outer core solidifies" (Pappas, 2023). Like the convection currents, this solidification also drives the circulation of the outer core, creating the magnetosphere. It's not yet understood what the observed change in pace could mean for the production of the magnetosphere. If this is just part of a cycle like Yang and Song think, it will likely be insignificant. Tkalčić thinks the cycle takes 20 to 30 years, and this shift is a normal occurrence (Hurst, 2023).

Concluding Thoughts on Misinterpreting Data

It's possible that these two studies could be connected to one another, and the results are being wrongly assessed. It could be that the inner core just doesn't have a smooth surface, and instead, it's a rough and fluctuating surface. Both of these discoveries might be evidence of the same thing. Lianxing Wen from Stony Brook University believes "the inner core has a shifting topography," which to him, this interpretation "best explains [the] observed temporal changes of seismic waves" (Pappas, 2023). The difference in the earthquakes could be attributed to a variety of different factors that don't necessarily correlate to a change in spin direction or speed. We may never truly know what's under the Earth's surface, but human curiosity continues the search forward; "because of its inaccessibility, this abyssal realm may forever elude explanation" (Andrews, 2023).

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(Kuthunur, 2023)

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7. The Ethics and Science of Gene Editing in Humans

Olivia Cohn '26

Advancements in genetic engineering and gene editing technologies have the potential to significantly improve human health and eradicate genetic disorders. However, as scientists delve deeper into this technology, its various implications raise important ethical questions that must be addressed as a society.

Genetic Engineering and Gene Editing

Gene engineering refers to the intentional manipulation of genetic material to alter the characteristics of an organism. The technology was developed in the 1970s, enabling scientists to manipulate DNA. More recently, gene editing has become a powerful tool that allows scientists to selectively add, delete, or modify genes with remarkable precision (Cho & Juengst, 2018). One of the most widely used gene-editing technologies is CRISPR-Cas9, which stands for Clustered Regularly Interspaced Short Palindromic Repeats. The CRISPR-Cas9 system is based on a naturally occurring defense mechanism used by bacteria to protect themselves against viruses. Cas9 is an enzyme that acts as molecular scissors, cutting DNA at a specific location determined by a guide RNA molecule. By designing a guide RNA that matches a specific sequence of DNA, scientists can direct the Cas9 enzyme to cut that sequence, enabling them to add, remove, or replace genetic material with great precision (Doudna & Charpentier, 2014).

Ethical Concerns

One of the main ethical concerns surrounding gene editing is the potential for its use in the creation of "designer babies". This would involve selecting desirable traits and altering them at the embryonic stage. While this may seem like a logical progression of technology, the potential repercussions of this practice raise significant ethical issues. It would result in a loss of diversity and could create a new era of eugenics, where only the genetically enhanced have access to opportunities and resources (National Academies of Sciences, Engineering, and Medicine, 2017).

Another ethical concern is the impact of gene editing on future generations. It is possible that some of the changes made in gene editing could have unintended consequences that could be passed down. Furthermore, as with any new technology, there is a risk that it could be used in a way that violates ethical standards. It is important to have policies and regulations in place to ensure that gene

editing is used in a responsible manner (Cho & Juengst, 2018).

Successes in Gene Editing

Despite the ethical concerns, gene editing has already been successful in treating genetic disorders. For example, it has been used to treat a form of blindness and to cure sickle cell disease (National Academies of Sciences, Engineering, and Medicine, 2017). Additionally, there is the potential to reduce the risk of certain diseases, such as cancer, by modifying the genes that predispose individuals to those conditions.

Conclusion

In conclusion, gene editing has the potential to significantly improve human health and eradicate genetic disorders. With responsible use, gene editing could change the world, and what it means to be a human. However, it is important to proceed with caution and consider the implications of this technology.

The Ship of Theseus riddle serves as an interesting analogy for the potential impact of gene editing. If the boat gets replaced with parts, is it still the same boat? Similarly, with gene editing, if a person's genetic makeup is significantly altered, are they still the same person?

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8. Debunking Conflicting Information about Saturated Fats: A Review

Leigh Foran '24

What is Saturated Fat?

Saturated fats are a type of lipid which contain only single bonds in the carbon chain. In these organic molecules, all excess carbon electrons are left bonded to hydrogen (Kahn). Many foods such as meat, butter, dairy products and foods contain saturated fats. They are also found in certain plant-derived foods including coconut oil and palm oil ("Saturated fat"). While healthier fats such as monounsaturated and polyunsaturated fats tend to be liquid at room temperature, saturated fats are typically solid (Manetti, 2022).

However, saturated fats are not as unhealthy as trans fats. Consuming trans fats leads to augmented LDL cholesterol and can even lower "good" HDL cholesterol (Mensick et al., 2003).

Health Impacts of Saturated Fat

Studies have shown that saturated fats may have many negative health impacts. Namely, the risk of cardiovascular disease has been shown to increase in people who consume higher levels of saturated fats. A study at Harvard University associated an 18% increased relative risk of coronary heart disease with a higher intake of the most commonly consumed major saturated fatty acids—lauric acid, myristic acid, palmitic acid, and stearic acid (Datz).

Commenting on this discovery, senior author Qi Sun, assistant professor in the Department of Nutrition, stated, "our findings strongly corroborate what the current USDA Dietary Guidelines recommend." They continued, "this includes reducing saturated fat intake to no more than 10% of total calories."

Walter Willett, a co-author of the study and professor of epidemiology and nutrition, added, "Replacing sources of saturated fat in our diets with unsaturated fats is one of the easiest ways to reduce our risk of heart disease." (Datz).

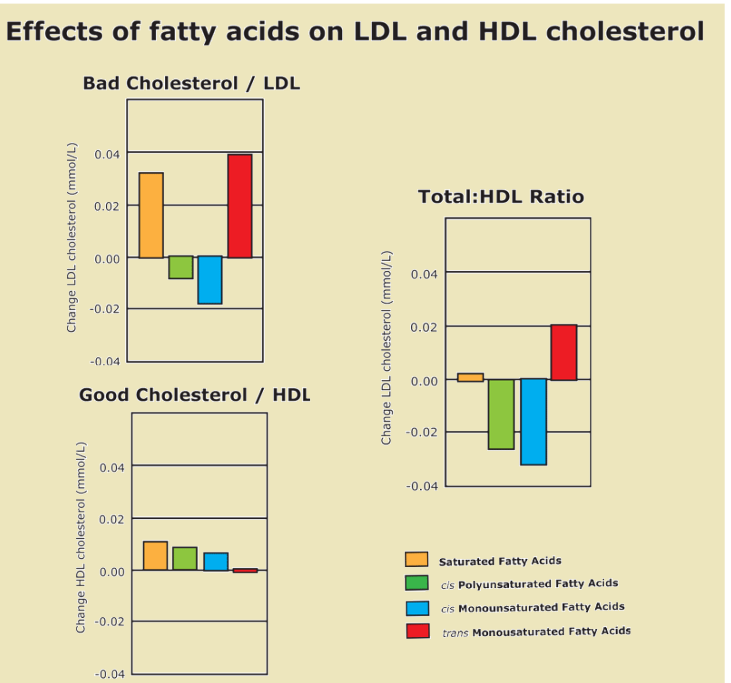
Furthermore, there has been certain evidence which suggests that saturated fats and high cholesterol levels may be linked to an increased risk of Alzheimer's disease or other diseases that cause dementia (Mayo Clinic Staff, 2023).

The Medical Debate

Since the publication of early American epidemiological studies, healthcare professionals have long vilified saturated fat. Citing its propensity to raise LDL cholesterol levels in the blood, nutritionists recommended that Americans limit their daily intake of these lipids (Cassiday, 2015). However, although initial epidemiological studies associated saturated fat intake with heart disease risk, following studies have failed to confirm the link. A paper published in the January 2015 edition of the British Medical Journal examined the data on fat and cardiovascular disease available to US and UK regulatory committees at the time. It revealed that the randomized controlled trials available back then did not provide sufficient evidence that cutting saturated fat intake reduces deaths from heart disease. The authors conclude that the "dietary advice not merely needs review; it should not have been introduced." (Harcombe et al., 2015).

Additionally, saturated fat raises HDL ("good") cholesterol levels, perhaps mitigating its effects on LDL cholesterol (Cassiday, 2015). Furthermore, exchanging saturated fats for a low-fat diet may lead to increased carbohydrate intake. This can actually raise heart disease risk compared with a higher-fat diet (Liu et al., 2017).

According to Nina Teicholz, author of The New York Times bestselling book The Big Fat Surprise: Why Butter, Meat & Cheese Belong in a Healthy Diet, in 1960 approximately 40% of calories in the American diet came from fats. Currently, total fats make up only 30% of total calories and saturated fat consumption has been reduced to about 11% (Wright and Wang, 2010). Yet obesity, heart disease, and diabetes continue to be problems, bringing into question the association between saturated fats and these health problems.



Fatty acid effects on LDL and HDL cholesterol are shown in FIG. 2. (Left) Trans fats and saturated fats raise the amount of LDL, or "bad," cholesterol in the serum. (Center) Fatty acids, both saturated and unsaturated, raise HDL, or "good," cholesterol. (Right) The ratio of total cholesterol to HDL declines when unsaturated fats are consumed, lowering the risk of heart disease. Saturated fats do not significantly increase or decrease this ratio, however trans fats do. (Cassiday, 2015)

Guidelines and Recommendations

The American Heart Association recommends aiming for dietary habits that achieve 5% to 6% of calories from saturated fat each day (American Heart Association editorial staff, 2021). For example, if a person consumes about 2,000 calories daily, no more than 120 of them should come from saturated fat. This translates to about 13 grams of saturated fat per day.

On the other hand, the 2020-2025 Dietary Guidelines for Americans state that people should get no more than 25% to 30% of their daily calories from fats. This includes limiting saturated fat to less than 10% of daily calories: for a 2,000-calorie diet, that is 200 calories or 20 grams of saturated fats a day (Manetti, 2022). As an example, even 1 tablespoon of butter contains 7 g of saturated fat (almost a third of the recommended daily allowance).

For those who suffer from heart disease or high cholesterol, health care providers recommend limiting saturated fat intake even more (Datz).

Alternatives

Medical professionals recommend switching to more wholesome, natural diets in order to reduce saturated fat intakes. This involves incorporating more fruits, vegetables, and whole grains into daily diets. Additionally, Americans can cut down on saturated fat by substituting unhealthy food with healthier options. This can include replacing foods high in saturated fats with foods that have polyunsaturated and monounsaturated fats.

Some examples may be replacing red meats with skinless chicken or fish a few days a week. Additionally, using canola or olive oil instead of butter and other solid fats can be a good option for cooking and baking. Finally, limiting the consumption of whole-fat dairy and instead using low-fat or nonfat milk, yogurt, and cheese can help as well.



(Saturated fat.)

Yet, just replacing saturated fat with refined carbohydrates such as sugary foods and drinks won't improve health (Manetti, 2022). Instead, switching to unsaturated fats and cooking with vegetable oils like rapeseed or sunflower oil may reduce risk of heart attack and stroke ("What does fat do," 2022).

Conclusion

Although there has been some debate about whether saturated fats should be viewed as truly unhealthy sources of nutrition, numerous studies have confirmed their negative health impacts. Dr. Frank Hu, professor of nutrition and epidemiology at Harvard University, encourages Americans to "dispel the notion that 'butter is back.'" He explained that "it is healthier to replace these fatty acids with unsaturated fats from vegetable oils, nuts, seeds, and seafood as well as high quality carbohydrates." (Datz).

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